PARIKSHA365 — STUDY NOTES

Chemistry — Common Book

SSC, RRB, State PSC, Banks (GA) — General Science

SSC RRB BANKS STATE PSC

v 2026-04

The promise. Read this book end-to-end (with the usual two re-reads + active recall on the embedded prompts) and you will be able to attempt every quiz question in the Pariksha365 pool for this subject with a strong fundamental grasp.

Examiner's Blueprint — What Chemistry Questions Are Really About

⚠ **EXAM ALERT**

Walk into any SSC CGL, IBPS PO or RRB exam and chemistry questions follow a brutal pattern: they are almost never about deep understanding. They are about recall — specific, well-targeted recall. Examiners love three things above all: **1. Everyday names vs chemical names** ("What is the chemical name of baking soda?") — this appears in almost every paper. **2. Alloy composition** ("Which metals make brass?") — expect at least one in every General Science section. **3. Discovery/inventor facts** ("Who invented nylon?") — steady 1-2 questions per paper in SSC exams. Everything else — pH values, activity series, industrial processes, polymers, indicators — shows up regularly but less predictably. This book is built around these three hot zones, then covers the rest in depth so you're never caught off guard.

Priority order for your study time:

1. Everyday chemical names table (Part U) — memorise every row
2. Alloys table (Part V) — know composition + use for top 15 alloys
3. Discovery/inventor dates (Part T) — the "who discovered X" questions
4. Acids, bases, pH, indicators (Part C-D) — 2-3 Q every paper
5. Activity series + metal reactions (Part E) — displacement questions
6. Polymers + their monomers (Part F) — SSC loves this
7. Industrial processes — Haber, Contact, Hall-Héroult (Part AA)
8. States of matter + periodic table trends (Parts A-B)
9. Organic basics — functional groups, fuels (Parts G-H)
10. Fertilisers, food chemistry, environmental (Parts Y, J, K)

PART A — MATTER AND ITS NATURE

Chapter A1 — States of Matter

Picture a ₹10 ice block on a summer day. In 20 minutes you watch it go solid → liquid → gas (steam). Same H_2O molecules throughout — only their motion and spacing change. This is the entire story of states of matter.

The Five States

- **Solid** — definite shape and volume; particles vibrate in fixed positions around a lattice point.
- **Liquid** — definite volume, no fixed shape; particles slide over each other.
- **Gas** — no fixed shape or volume; particles move freely and fill any container.
- **Plasma** — ionised gas where electrons are stripped from atoms. Found in the Sun, stars, lightning, fluorescent tubes, neon signs.
- **Bose-Einstein Condensate (BEC)** — occurs at temperatures near absolute zero (−273 °C). Particles lose individual identities and behave as a single "super-atom." Predicted by Satyendra Nath Bose and Albert Einstein; first created in a lab in 1995.

✔ KEY FACT

The 5th state of matter is Bose-Einstein Condensate — NOT plasma. Examiners specifically ask this. Plasma is common around us (the Sun, tube lights, lightning); BEC is exotic and lab-made.

Phase Change Names (all six)

From	To	Name
Solid	Liquid	Melting (Fusion)
Liquid	Solid	Freezing (Solidification)
Liquid	Gas	Evaporation / Boiling
Gas	Liquid	Condensation
Solid	Gas	**Sublimation**
Gas	Solid	**Deposition** (reverse sublimation)

⚠ EXAM ALERT

Sublimation is a high-frequency topic. Substances that sublime directly from solid to gas without becoming liquid: **dry ice (CO_2), iodine, naphthalene (moth balls), camphor, ammonium chloride**. Frost forming on a cold window is deposition (reverse sublimation).

Chapter A2 — Atomic Structure

The Atomic Model: A History Worth Knowing

Examiners love "who discovered what?" questions in atomic structure. Learn this sequence cold.

Scientist	Year	Contribution
John Dalton	1808	Atoms are indivisible; each element has characteristic atoms
J.J. Thomson	1897	Discovered the **electron**; proposed plum-pudding model
Ernest Rutherford	1911	Alpha-scattering experiment → nucleus discovered; "mostly empty space"
Henry Moseley	1913	Measured atomic number using X-rays; fixed the periodic table order
Niels Bohr	1913	Quantised electron orbits; explained hydrogen spectrum
James Chadwick	1932	Discovered the **neutron**
Quantum model	1926+	Electrons occupy probability clouds (orbitals), not fixed paths

✔ KEY FACT

Three discoveries that examiners repeat: **electron → J.J. Thomson (1897), proton → Rutherford (1919), neutron → Chadwick (1932)**. The heaviest sub-atomic particle is the neutron (just slightly heavier than the proton).

Key Definitions

- **Atomic number (Z)** = number of protons = number of electrons in a neutral atom.
- **Mass number (A)** = protons + neutrons.
- **Isotopes** — same Z, different A. Example: Hydrogen has three isotopes: ^{1}H (protium), ^{2}H (deuterium), ^{3}H (tritium). Heavy water uses deuterium: D_2O.
- **Isobars** — same A, different Z. Example: ^{40}Ar, ^{40}K, ^{40}Ca.
- **Isotones** — same number of neutrons, different Z.

◎ MNEMONIC

To remember isotopes vs isobars: **"Same Z = IsoTopes (T = Top = atomic number on top in notation)"** and **"Same A = IsobArs (A = same mass number)"**.

PART B — THE PERIODIC TABLE

Chapter B1 — From Mendeleev to Moseley

In 1869, Dmitri Mendeleev arranged 63 known elements by atomic weight and noticed repeating patterns in behaviour — he called this the **periodic law**. More importantly, he was bold enough to leave gaps for elements that didn't yet exist. He predicted eka-aluminium (later found: Gallium), eka-silicon (Germanium), and eka-boron (Scandium). All three were discovered within 15 years, confirming the table.

The problem: Mendeleev's table had anomalies — argon (Ar, atomic weight 39.9) came before potassium (K, 39.1) in weight order but clearly belonged in the noble gas column. **Henry Moseley (1913)** solved this by re-ordering on atomic number, not atomic weight. All anomalies vanished.

✔ KEY FACT

The modern periodic table has **7 periods (rows)**, **18 groups (columns)**, **118 elements**, and is divided into **4 blocks: s, p, d, f**. The last four elements (113, 115, 117, 118 — Nihonium, Moscovium, Tennessine, Oganesson) were officially added in **2016**.

The Four Blocks

- **s-block** (Groups 1–2): Alkali metals (Li, Na, K, Rb, Cs, Fr) and alkaline earth metals (Be, Mg, Ca, Sr, Ba, Ra). Soft, reactive, low melting points.
- **p-block** (Groups 13–18): Contains the most variety — metals, metalloids, non-metals, noble gases. Includes carbon, nitrogen, halogens.
- **d-block** (Groups 3–12): Transition metals. Iron, copper, gold, silver, platinum. Hard, high melting points, variable oxidation states.
- **f-block** (below main table): Lanthanides (4f, rare earths) and Actinides (5f, radioactive). All actinides beyond uranium are man-made.

Important Groups — Learn These Families

Group	Family	Members	Character
1	Alkali metals	Li, Na, K, Rb, Cs, Fr	1 valence e^-; very reactive; soft
2	Alkaline earth metals	Be, Mg, Ca, Sr, Ba, Ra	2 valence e^-; less reactive than Group 1
13	Boron family	B, Al, Ga, In, Tl	3 valence e^-
14	Carbon family	C, Si, Ge, Sn, Pb	4 valence e^-
15	Pnictogens	N, P, As, Sb, Bi	5 valence e^-
16	Chalcogens	O, S, Se, Te, Po	6 valence e^-

Group	Family	Members	Character
17	Halogens	F, Cl, Br, I, At	7 valence e^-; most reactive non-metals
18	Noble gases	He, Ne, Ar, Kr, Xe, Rn	Full outer shell; inert

◎ MNEMONIC

Noble gases: **"He Never Argues; Keeps Xtra Rest"** — He, Ne, Ar, Kr, Xe, Rn. Halogens: **"Fierce Clowns Bring Interesting Acts"** — F, Cl, Br, I, At.

Periodic Trends — The Direction Rules

Trend	Across a period (left → right)	Down a group
Atomic size	Decreases	Increases
Ionisation energy	Increases	Decreases
Electron affinity	Increases	Decreases
Electronegativity	Increases	Decreases
Metallic character	Decreases	Increases

⚠ EXAM ALERT

The most electronegative element is **Fluorine (F), 4.0 on Pauling scale**. The most reactive non-metal is Fluorine. The most reactive metal is **Caesium (Cs)** — sometimes Francium is given as the answer, but Cs is more commonly tested. The lightest metal is **Lithium (Li)**. The only liquid metal at room temperature is **Mercury (Hg)** — Gallium is also liquid near room temperature (melts at 29.8 °C) and appears as a trap option.

◎ MNEMONIC

Electronegativity decreasing order: **F > O > N > Cl > Br > I**. Memory: **"French Ocean Navy Cleans Burmese Islands."**

PART C — CHEMICAL BONDING AND REACTIONS

Chapter C1 — How Atoms Bond Together

There are three main types of chemical bond. Understanding which type forms tells you almost everything about a compound's properties.

Ionic Bond

Formed by **transfer of electrons** — one atom gives, another takes. Results in oppositely charged ions attracted to each other. Typically between a metal and a non-metal.

Examples: $NaCl$ (table salt), MgO (magnesia), CaF_2 (fluorspar).

Properties: High melting point, conducts electricity when dissolved in water or melted, brittle.

Covalent Bond

Formed by **sharing of electrons** between two non-metals. Can be single (one shared pair), double (two shared pairs), or triple (three shared pairs).

Examples: H_2O (water), CH_4 (methane), O_2 (oxygen — double bond), N_2 (nitrogen — triple bond).

Metallic Bond

In metals, valence electrons form a "sea" that flows freely around positive metal cores. This is why metals conduct electricity and heat, are malleable, and are lustrous.

Dative (Coordinate) Bond

Both electrons in the shared pair come from the **same atom**. Examples: NH_4^+ (ammonium ion), H_3O^+ (hydronium ion).

Hydrogen Bond

A weak intermolecular attraction that forms when hydrogen is bonded to a very electronegative atom — **F, O, or N** only.

✔ KEY FACT

Hydrogen bonding explains why water is liquid at room temperature (small molecule but high boiling point), why ice floats on water (hydrogen bonds form an open hexagonal lattice in ice → lower density than liquid water), and how DNA bases pair (A-T with 2 H-bonds, G-C with 3 H-bonds).

Chapter C2 — Types of Chemical Reactions

Every chemical change in a competitive exam question falls into one of these categories. Know them by name and example.

Type	Pattern	Example
Combination	A + B → AB	$CaO + H_2O \rightarrow Ca(OH)_2$
Decomposition	AB → A + B	$CaCO_3 \rightarrow CaO + CO_2$
Displacement	A + BC → AC + B	$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
Double displacement	AB + CD → AD + CB	$AgNO_3 + NaCl \rightarrow AgCl\downarrow + NaNO_3$
Redox	electron transfer	$Fe + O_2 \rightarrow Fe_2O_3$ (rusting)
Combustion	fuel + O_2 → CO_2 + H_2O + heat	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
Exothermic	releases heat	neutralisation, combustion
Endothermic	absorbs heat	photosynthesis, thermal decomposition

◎ MNEMONIC

Oxidation and reduction — **OIL RIG**: Oxidation Is Loss (of electrons), Reduction Is Gain (of electrons). In the reaction $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$, Zn loses electrons (oxidised), Cu^{2+} gains them (reduced). Zinc is the reducing agent; Cu^{2+} is the oxidising agent.

× COMMON TRAP

Students confuse "oxidation" with "gaining oxygen" — that is only the old, incomplete definition. The modern definition is about electron loss/gain. Rusting is a redox reaction (iron loses electrons), not just "combining with oxygen."

PART D — ACIDS, BASES, SALTS

Chapter D1 — Three Theories of Acids and Bases

Theory	Acid	Base	Limitation
Arrhenius (1884)	Gives H^+ in water	Gives OH^- in water	Only for water solutions
Brønsted-Lowry (1923)	H^+ donor	H^+ acceptor	Needs H^+ transfer
Lewis (1923)	Electron-pair acceptor	Electron-pair donor	Most general; covers all

Lewis theory is the broadest. $AlCl_3$ is a Lewis acid (accepts electrons) even though it has no H^+ to give.

Chapter D2 — The pH Scale

$pH = -\log[H^+]$. Think of pH as a 0-to-14 ruler where 0 is maximally acidic and 14 is maximally basic.

Substance	Approximate pH
Battery acid (H_2SO_4)	0
Stomach acid (gastric juice)	1.5–2
Lemon juice	2–3
Vinegar	2.5–3.5
Soft drinks (cola)	3
Tomato juice	4
Black coffee	5
Milk	6.5–6.8
Pure water	7
Blood	7.35–7.45
Sea water	8
Baking soda solution	9
Ammonia (household)	11
Bleach	13
Caustic soda (NaOH)	14

⚠ **EXAM ALERT**

Blood pH is 7.35–7.45 — slightly alkaline. A drop below 7.35 causes acidosis; above 7.45 causes alkalosis, both life-threatening. Gastric juice at pH 1.5–2 is almost as acidic as battery acid — yet your stomach lining survives because of a mucus coating.

Chapter D3 — Indicators (Learn the Colour Changes)

Indicator	In Acid	In Neutral	In Base
Litmus	Red	Purple	Blue
Phenolphthalein	Colourless	Colourless	Pink
Methyl orange	Red	Orange	Yellow
Methyl red	Red	Orange	Yellow
Bromothymol blue	Yellow	Green	Blue
Universal indicator	Red→Orange	Green	Blue→Violet
Turmeric (natural)	Yellow	Yellow	Red-brown

◎ **MNEMONIC**

Litmus: **A-Red, B-Blue** — Acids turn litmus Red, Bases turn it Blue. Phenolphthalein: **colourless in acid, PiNk in base** — the P is for Pink in base.

Chapter D4 — Common Acids and Their Uses

Acid	Formula	Common name / use
Hydrochloric acid	HCl	Spirit of salt; gastric juice; metal cleaning; descaler
Sulphuric acid	H_2SO_4	Oil of vitriol; "king of chemicals"; car batteries; fertilisers
Nitric acid	HNO_3	Aqua fortis; explosives; fertilisers
Acetic acid	CH_3COOH	Vinegar; food preservative
Citric acid	$C_6H_8O_7$	Lemons, oranges; sour taste
Tartaric acid	$C_4H_6O_6$	Tamarind; baking powder ingredient
Oxalic acid	$H_2C_2O_4$	Tomatoes, spinach; stain remover
Lactic acid	$CH_3CH(OH)COOH$	Curd; sour milk
Formic acid	$HCOOH$	Ant sting; bee venom

✔ KEY FACT

Aqua regia = 3 parts HCl + 1 part HNO_3 (by volume). It dissolves gold and platinum — the only common mixture that can. This question appears in almost every competitive exam's chemistry section.

Chapter D5 — Common Bases and Their Uses

Base	Formula	Common name / use
Sodium hydroxide	NaOH	Caustic soda; soap-making, paper
Potassium hydroxide	KOH	Caustic potash; soft soap, batteries
Calcium hydroxide	$Ca(OH)_2$	Slaked lime; whitewash, mortar
Magnesium hydroxide	$Mg(OH)_2$	Milk of magnesia; antacid
Ammonium hydroxide	NH_4OH	Household cleaner

Chapter D6 — Salts and Their pH

Salt	Acid + Base used	pH
Sodium chloride (NaCl)	Strong + Strong	Neutral (7)
Ammonium chloride (NH_4Cl)	Weak base + Strong acid	Acidic (<7)
Sodium carbonate (Na_2CO_3)	Strong base + Weak acid	Basic (>7)
Sodium bicarbonate ($NaHCO_3$)	Strong base + Weak acid	Slightly basic (8.3)

⚠ EXAM ALERT

Baking soda is $NaHCO_3$ (sodium bicarbonate). It is mildly basic (pH ~8.3). When you eat something acidic, taking baking soda neutralises the acidity in your stomach. Also used in fire extinguishers — it decomposes on heating to release CO_2 which smothers fires.

PART E — METALS AND THE ACTIVITY SERIES

Chapter E1 — Properties of Metals

General properties: Lustrous, malleable, ductile, good conductors of heat and electricity, typically solid at room temperature with high melting points.

Important exceptions — examiners love these:

- **Mercury (Hg)** — only metal liquid at room temperature. (Gallium also melts near room temperature at 29.8 °C — a common trap.)
- **Sodium (Na) and Potassium (K)** — metals but so soft you can cut them with a knife; stored in kerosene oil to prevent reaction with air/water.
- **Gold (Au)** — most malleable and ductile metal. 1 gram of gold can be drawn into 2.4 km of wire.
- **Silver (Ag)** — best conductor of electricity. We use copper in wiring only because silver is too expensive.
- **Tungsten (W)** — highest melting point among all metals (3422 °C). Used in light bulb filaments.
- **Lithium (Li)** — lightest metal (density 0.534 g/cm³).
- **Osmium (Os)** — densest naturally occurring metal.

Chapter E2 — Activity Series (Reactivity Series)

This is one of the most frequently tested topics in competitive chemistry. Memorise the order.

Most reactive → Least reactive:

K > Na > Ca > Mg > Al > Zn > Fe > Sn > Pb > **[H]** > Cu > Hg > Ag > Au > Pt

◎ MNEMONIC

"Please Stop Calling Me A Zebra, I Stay Planted Here, Cause He's Still At Practice" P = Potassium (K), S = Sodium (Na), C = Calcium (Ca), M = Magnesium (Mg), A = Aluminium (Al), Z = Zinc (Zn), I = Iron (Fe), S = Tin (Sn), P = Lead (Pb), H = Hydrogen, C = Copper (Cu), H = Mercury (Hg), S = Silver (Ag), A = Gold (Au), P = Platinum (Pt)

What the Series Tells You

- Metals above hydrogen in the series can displace H from acids. Metals below hydrogen (Cu, Hg, Ag, Au, Pt) cannot — they do not react with dilute acids.
- A metal higher in the series displaces a lower one from its salt solution. This is why $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ (zinc displaces copper because Zn is higher than Cu).
- K, Na react with cold water violently. Ca and Mg react with hot water. Fe reacts with steam.

× COMMON TRAP

"Most reactive metal" — the answer examiners want is Caesium (Cs) or Potassium (K) depending on how the question is framed. Francium (Fr) is also in Group 1 and is technically most reactive, but it is synthetic and radioactive, so practical exams usually mean Cs or K.

Chapter E3 — Extraction of Metals

Metal	Ore (common)	Extraction method
Aluminium	Bauxite ($Al_2O_3 \cdot 2H_2O$)	Bayer process (concentration) + Hall-Héroult (electrolysis in molten cryolite)
Iron	Haematite (Fe_2O_3)	Blast furnace with coke
Copper	Copper pyrites ($CuFeS_2$)	Roasting + smelting + Bessemer process
Zinc	Zinc blende (ZnS)	Roasting + carbon reduction
Lead	Galena (PbS)	Roasting + reduction
Mercury	Cinnabar (HgS)	Direct heating — Hg vapour condenses
Sodium	Rock salt ($NaCl$)	Down's process (electrolysis of molten NaCl)
Silver	Argentite (Ag_2S)	Cyanide leaching
Gold	Native gold	Cyanidation with NaCN solution
Tin	Cassiterite (SnO_2)	Carbon reduction

✔ KEY FACT

Hall-Héroult process (1886): Electrolysis of alumina (Al_2O_3) dissolved in molten cryolite (Na_3AlF_6). This is how all commercial aluminium is made. Cryolite is the solvent that lowers the melting point from 2050 °C to around 950 °C, making the process practical.

Chapter E4 — Corrosion and Prevention

- **Rusting (iron):** $Fe + O_2 + H_2O \rightarrow$ hydrated iron oxide ($Fe_2O_3 \cdot xH_2O$) — the red-brown flaky coating. Both oxygen AND water are needed; iron does not rust in dry air or pure water alone.
- **Tarnishing (silver):** $Ag + H_2S \rightarrow Ag_2S$ (black coating).
- **Patina (copper/bronze):** Green coating on copper statues = basic copper carbonate.

Prevention methods: - Galvanising — coating iron with zinc. Even if the Zn coat is scratched, Zn corrodes preferentially (sacrificial anode). - Tinning — coating with tin (food cans). If scratched, iron corrodes faster. - Electroplating — chrome, nickel. - Alloying — stainless steel (Cr + Ni) is naturally corrosion-resistant. - Painting, oiling, varnishing.

⚠ EXAM ALERT

Galvanising uses zinc as a protective coat. Zinc is above iron in the activity series, so zinc acts as a sacrificial anode — it oxidises preferentially, protecting the iron. This is cathodic protection. Tinning uses tin (below iron in the series), so if the tin coat is scratched, iron corrodes faster.

PART F — ORGANIC CHEMISTRY

Chapter F1 — Why Carbon Is Special

Carbon forms 4 bonds (tetravalent), chains indefinitely with itself (catenation), and forms stable double and triple bonds. Result: over 10 million known organic compounds versus about 500,000 for all other elements combined. No other element comes close.

Chapter F2 — Hydrocarbons — The Three Series

Alkanes — C_nH_{2n+2} (single bonds, saturated)

Methane (CH_4), Ethane (C_2H_6), Propane (C_3H_8), Butane (C_4H_{10}), Pentane (C_5H_{12}), Hexane (C_6H_{14}), Heptane (C_7H_{16}), Octane (C_8H_{18}), Nonane (C_9H_{20}), Decane ($C_{10}H_{22}$).

◎ **MNEMONIC**

First 10 alkanes: **"Me Eats Peanut Butter, Pretty Happily; Hears Only Nice Decrees"** — Methane, Ethane, Propane, Butane, Pentane, Hexane, Heptane, Octane, Nonane, Decane.

Alkenes — C_nH_{2n} (one double bond, unsaturated)

Ethene (C_2H_4), Propene (C_3H_6), Butene (C_4H_8)...

Ethene (ethylene) is the plant hormone that triggers fruit ripening. This is why a ripe banana in a fruit bowl makes other fruits ripen faster.

Alkynes — C_nH_{2n-2} (one triple bond, unsaturated)

Ethyne (acetylene, C_2H_2) is the most important. It burns at over 3000 °C with oxygen — used in welding (oxyacetylene torch).

Chapter F3 — Functional Groups

Group	Formula	Class	Key Example
Hydroxyl	−OH	Alcohol	Ethanol (C_2H_5OH) — drinking alcohol
Aldehyde	−CHO	Aldehyde	Formaldehyde (HCHO) — preservative
Ketone	>C=O	Ketone	Acetone (CH_3COCH_3) — nail polish remover
Carboxyl	−COOH	Carboxylic acid	Acetic acid (CH_3COOH) — vinegar
Ester	−COO−	Ester	Ethyl acetate — nail polish remover solvent
Amine	$-NH_2$	Amine	Methylamine

Group	Formula	Class	Key Example
Halide	−X	Haloalkane	Chloroform ($CHCl_3$)

Chapter F4 — Fuels

⚠ EXAM ALERT

Fuel questions appear in almost every competitive exam. Know these three: **LPG = butane + propane; CNG = mainly methane; biogas = ~60% methane + CO_2**. The smell of LPG is due to ethyl mercaptan — LPG itself is odourless.

Composition of Common Fuels

Fuel	Main composition	Notes
LPG	Propane (C_3H_8) + Butane (C_4H_{10})	Ethyl mercaptan added for smell detection
CNG	Methane (~85–95%) + small amounts of ethane, propane	Cleaner than petrol/diesel
Petrol (gasoline)	Hydrocarbons C_5–C_{12} (mainly octanes)	Octane number = % iso-octane
Diesel	Hydrocarbons C_{12}–C_{20}	Cetane number = ignition quality
Kerosene	C_{12}–C_{15}	Jet fuel, household
Biogas (gobar gas)	~60% CH_4 + CO_2 + traces H_2S	From animal waste fermentation
Coal gas	H_2 (~50%) + CH_4 (~30%) + CO	From destructive distillation of coal
Water gas	CO + H_2 (1:1)	Coke + steam at high temp
Producer gas	CO + N_2 (~70%)	Coke + air; cheaper but less energy
Natural gas	~85% methane + ethane + propane	Found with petroleum deposits

Calorific Values — Energy per kg

Fuel	kJ/kg (approximate)
Hydrogen	150,000
Methane (CNG)	55,000
LPG	50,000
Petrol / Diesel / Kerosene	45,000

Fuel	kJ/kg (approximate)
Coal (anthracite)	33,000
Wood	17,000
Cow-dung cake	6,000

Hydrogen has by far the highest calorific value — nearly 3× petrol. This is why it is called the "fuel of the future."

Coal Types by Carbon Content

Type	% Carbon	Quality
Peat	50–60%	Lowest; youngest
Lignite (brown coal)	60–70%	Low
Bituminous	78–87%	Most widely used
Anthracite	87–95%	Highest CV; smokeless

PART G — POLYMERS AND FIBRES

Chapter G1 — Key Concepts

- **Monomer** — small repeating unit.
- **Polymer** — long chain of monomers linked together.
- **Addition polymerisation** — monomers join without losing atoms (e.g., ethene → polythene).
- **Condensation polymerisation** — monomers join with release of a small molecule like water or HCl (e.g., nylon, polyester, bakelite).
- **Thermoplastic** — softens on heating, can be remoulded. (PVC, polythene, polypropylene)
- **Thermosetting** — once set, cannot be melted again. (Bakelite, melamine, epoxy resin)

Chapter G2 — The Important Polymers Table

Polymer	Monomer(s)	Type	Common Uses
Polythene (LDPE/HDPE)	Ethene	Addition, thermoplastic	Bags, bottles, pipes
Polypropylene (PP)	Propene	Addition, thermoplastic	Ropes, carpets, food containers
PVC	Vinyl chloride	Addition, thermoplastic	Pipes, raincoats, insulation
Polystyrene	Styrene	Addition, thermoplastic	Disposable cups, packaging foam
Teflon (PTFE)	Tetrafluoroethylene	Addition, thermoplastic	Non-stick cookware, lab seals
Plexiglas (PMMA)	Methyl methacrylate	Addition	Unbreakable glass, contact lenses
Nylon-6,6	Adipic acid + hexamethylenediamine	Condensation	Ropes, parachutes, stockings
Nylon-6	Caprolactam	Condensation	Carpets, fabric
Terylene / Dacron / PET	Ethylene glycol + terephthalic acid	Condensation	Polyester fabric, bottles, audio tape
Bakelite	Phenol + formaldehyde	Condensation, thermosetting	Electrical switches, handles
Melamine	Melamine + formaldehyde	Condensation, thermosetting	Crockery, fire-retardant fabrics

Polymer	Monomer(s)	Type	Common Uses
Buna-S (SBR)	Butadiene + styrene	Addition, synthetic rubber	Tyres, footwear
Buna-N (NBR)	Butadiene + acrylonitrile	Addition, synthetic rubber	Oil-resistant hoses, gaskets
Neoprene	Chloroprene	Addition, synthetic rubber	Wetsuits, weather-resistant seals
Natural rubber	Isoprene (polyisoprene)	Natural	Tyres (after vulcanisation)
Cellulose	β-glucose	Natural	Cotton, paper, wood
Starch	α-glucose	Natural	Food storage in plants
Rayon	Cellulose (treated)	Semi-synthetic	"Artificial silk"

⚠ EXAM ALERT

Three "first" questions that appear repeatedly: **First synthetic plastic = Bakelite (1907, Leo Baekeland)**. **First fully synthetic fibre = Nylon (1935, Wallace Carothers, DuPont)**. **Non-stick pan coating = Teflon (PTFE, Roy Plunkett, DuPont, 1938)**. Commit all three discoverers and years to memory.

Chapter G3 — Vulcanisation of Rubber

Natural rubber (polyisoprene) is sticky, weak, and melts in summer heat. In 1839, Charles Goodyear discovered that heating raw rubber with sulphur creates cross-links between the polymer chains. This process — vulcanisation — makes rubber harder, stronger, elastic, and heat-stable. All tyre rubber is vulcanised.

PART H — EVERYDAY CHEMISTRY

Chapter H1 — Food Science

- **Fermentation** — yeast converts sugars into ethanol + CO_2 without oxygen (anaerobic). Basis of bread (CO_2 makes dough rise), beer, wine, idli.
- **Pasteurisation** — heating milk to 72 °C for 15 seconds, then rapid cooling. Kills pathogenic bacteria without changing flavour significantly. Developed by Louis Pasteur.
- **Hydrogenation** — liquid vegetable oils + H_2 gas (Ni catalyst) → solid fats (Vanaspati ghee). Alkenes become alkanes. Produces trans fats as a side product — these are associated with heart disease.
- **Iodised salt** — KIO_3 (potassium iodate) added to common salt to prevent iodine deficiency diseases, especially goitre.
- **Baking soda ($NaHCO_3$)** — on heating or in contact with acid, releases CO_2 which makes batter rise. Baking powder = $NaHCO_3$ + tartaric acid + starch.

Food Adulterants — Tested Questions

Food	Common adulterant	Detection test
Milk	Water, starch	Lactometer (for water); iodine turns blue (for starch)
Ghee	Vanaspati (vegetable fat)	Butyro-refractometer; Baudouin test
Honey	Sugar syrup	Fiehe's test (resorcinol + HCl)
Turmeric	Metanil yellow	Concentrated HCl → pink/red colour
Chilli powder	Brick powder	Water — brick powder settles
Mustard oil	Argemone oil	HNO_3 turns yellow (sanguinarine test)
Pepper	Papaya seeds	Water — papaya seeds are lighter, float

Chapter H2 — Soaps, Detergents, Water

Soap

Soap = sodium salt (or potassium salt) of a long-chain fatty acid. Made by saponification — heating fat/oil with NaOH. The soap molecule has a long hydrophobic tail (dissolves in oil) and a hydrophilic ionic head (dissolves in water). This is how it removes grease: the tails embed in the oil droplet, the heads stick out into water.

Soap does NOT work well in hard water. Hard water contains dissolved Ca^{2+} and Mg^{2+} ions. These react with soap to form insoluble scum (calcium stearate), which wastes soap and leaves a film.

Detergent

Synthetic detergents (alkyl benzene sulphonates) work in hard water because their calcium/magnesium salts are soluble. Newer detergents are biodegradable; older ones (branched-chain alkyl sulphonates) were not.

Hard Water

- **Temporary hardness** — caused by dissolved $Ca(HCO_3)_2$ and $Mg(HCO_3)_2$. Removed by boiling (bicarbonates decompose to insoluble carbonates that precipitate).
- **Permanent hardness** — caused by $CaSO_4$, $MgSO_4$, $CaCl_2$. Not removed by boiling. Removed by ion-exchange resins (zeolite) or adding Na_2CO_3 (washing soda).

Chapter H3 — Medicines

Drug class	What it does	Examples
Analgesic (painkiller)	Reduces pain	Aspirin, paracetamol, ibuprofen
Antipyretic	Reduces fever	Paracetamol (also analgesic)
Antibiotic	Kills bacteria	Penicillin, amoxicillin, streptomycin, tetracycline
Antiseptic	Kills microbes on living tissue	Dettol, iodine, hydrogen peroxide, savlon
Disinfectant	Kills microbes on non-living surfaces	Phenol, formalin, bleaching powder
Antacid	Neutralises stomach acid	Sodium bicarbonate, milk of magnesia, gelusil
Antihistamine	Treats allergic reactions	Cetirizine, chlorpheniramine
Antimalarial	Treats malaria	Chloroquine, quinine, artemisinin
Antitubercular	Treats TB	Rifampicin, isoniazid, pyrazinamide, ethambutol
Tranquiliser	Reduces anxiety	Diazepam (Valium), equanil

✓ KEY FACT

Penicillin — discovered by Alexander Fleming in **1928** when he noticed the mould Penicillium notatum killing bacteria in a petri dish. This remains one of the most frequently tested discovery facts in chemistry/general science.

PART I — ENVIRONMENTAL CHEMISTRY

Chapter I1 — Air Pollution

Pollutant	Source	Effect
CO (carbon monoxide)	Incomplete combustion, vehicle exhaust	Binds haemoglobin 240× stronger than O_2 → suffocation
SO_2 (sulphur dioxide)	Coal combustion → combines with rain → H_2SO_4	Acid rain; respiratory irritant
NO_x (nitrogen oxides)	Vehicle exhaust, thunderstorms	Photochemical smog; acid rain
$PM_{2.5}$ / PM_{10}	Dust, smoke, vehicle exhaust	Deep lung penetration; respiratory disease
Ground-level O_3	Photochemical reaction of NO_x + VOCs in sunlight	Lung irritant; reduces crop yields
Greenhouse gases	Various sources	Global warming

Greenhouse Gases — Global Warming Potential

Gas	GWP (relative to CO_2)
CO_2	1 (reference)
CH_4 (methane)	28–36
N_2O	298
HFCs	Hundreds to thousands
SF_6	23,500

⚠ **EXAM ALERT**

The main greenhouse gas by volume is CO_2. But the strongest greenhouse gas by GWP is SF_6 (sulphur hexafluoride). Examiners ask both. "Most abundant greenhouse gas in the atmosphere" = **water vapour (H_2O)** — but examiners almost always mean CO_2 when they ask this, as CO_2 is the main anthropogenic one.

Chapter I2 — Ozone Layer

- Ozone (O_3) in the stratosphere (20–30 km altitude) absorbs harmful UV-B and UV-C radiation from the Sun.

- CFCs (chlorofluorocarbons) — used in refrigerants and aerosols — decompose in the stratosphere and release chlorine atoms that catalytically destroy ozone.
- The ozone hole over Antarctica was discovered in 1985.
- **Montreal Protocol (1987)** — international treaty phasing out CFCs. Most successful environmental treaty.
- **Kigali Amendment (2016)** — extends Montreal Protocol to phase out HFCs (hydrofluorocarbons), which replaced CFCs but are potent greenhouse gases.

Chapter I3 — Water Pollution

- **BOD (Biological Oxygen Demand)** — amount of oxygen needed by bacteria to decompose organic matter in water. Higher BOD = more pollution.
- **DO (Dissolved Oxygen)** — fish need at least 4 mg/L. Heavily polluted water can have DO near zero.
- **Eutrophication** — excess nutrients (nitrates, phosphates from fertiliser runoff) → algal bloom → algae die → bacteria decompose them using up all oxygen → fish die.
- **Fluorosis** — fluoride >1.5 ppm in drinking water → mottled teeth, skeletal fluorosis.
- **Blue baby syndrome (Methaemoglobinaemia)** — high nitrates in well water → converted to nitrites by gut bacteria → blocks oxygen in infant blood.

Famous Pollution Disasters

Disaster	Cause	Location
Minamata disease	Mercury (Hg) poisoning from industrial discharge	Minamata Bay, Japan (1950s)
Itai-Itai disease	Cadmium (Cd) poisoning from mining discharge	Jinzu River, Japan (1950s)
Bhopal gas tragedy	MIC (methyl isocyanate) gas leak	Bhopal, India, 1984

PART T — DISCOVERIES AND INVENTORS IN CHEMISTRY

This section is pure exam gold — learn the discoverer and year for each entry.

Discovery / Invention	Person	Year
Atomic theory	John Dalton	1808
Electron	J.J. Thomson	1897
Radioactivity	Henri Becquerel	1896
Polonium + Radium	Marie & Pierre Curie	1898
Alpha, beta particles	Ernest Rutherford	1899
Nuclear model of atom	Ernest Rutherford	1911
Proton	Ernest Rutherford	1919
Neutron	James Chadwick	1932
Modern periodic table (by atomic number)	Henry Moseley	1913
Periodic law (by atomic weight)	Dmitri Mendeleev	1869
Synthesis of urea from inorganic compounds	Friedrich Wöhler	1828
Hydrogen	Henry Cavendish	1766
Oxygen	Joseph Priestley + Carl Scheele	1774
Nitrogen	Daniel Rutherford	1772
Vulcanisation of rubber	Charles Goodyear	1839
First synthetic dye (mauveine)	William Perkin	1856
Dynamite	Alfred Nobel	1867
Rayon (artificial silk)	Hilaire de Chardonnet	1884
Penicillin	Alexander Fleming	1928
Streptomycin (first anti-TB antibiotic)	Selman Waksman	1943
Bakelite (first true synthetic plastic)	Leo Baekeland	1907
Polythene	Eric Fawcett + Reginald Gibson (ICI)	1933
Nylon (first fully synthetic fibre)	Wallace Carothers, DuPont	1935

Discovery / Invention	Person	Year
Teflon (PTFE)	Roy Plunkett, DuPont	1938
Haber process (ammonia synthesis)	Fritz Haber + Carl Bosch	1909–1913
Contact process (sulphuric acid)	Peregrine Phillips	1831
Ostwald process (nitric acid)	Wilhelm Ostwald	1902
Hall-Héroult process (aluminium)	Charles Hall + Paul Héroult	1886
DDT (use as insecticide)	Paul Müller	1939
CFC refrigerants	Thomas Midgley Jr	1928
TNT	Julius Wilbrand	1863

PART U — THE ESSENTIAL CHEMICAL NAMES TABLE

This is the table you must memorise completely. Chemical name ↔ everyday name questions appear in every competitive exam paper.

Everyday Name	Chemical Name	Formula
Common salt	Sodium chloride	$NaCl$
Baking soda	Sodium bicarbonate	$NaHCO_3$
Washing soda	Sodium carbonate decahydrate	$Na_2CO_3 \cdot 10H_2O$
Caustic soda	Sodium hydroxide	$NaOH$
Caustic potash	Potassium hydroxide	KOH
Quicklime	Calcium oxide	CaO
Slaked lime	Calcium hydroxide	$Ca(OH)_2$
Limestone / Marble / Chalk	Calcium carbonate	$CaCO_3$
Plaster of Paris (POP)	Calcium sulphate hemihydrate	$CaSO_4 \cdot ½H_2O$
Gypsum	Calcium sulphate dihydrate	$CaSO_4 \cdot 2H_2O$
Bleaching powder	Calcium hypochlorite (oxychloride)	$CaOCl_2$
Hypo (photography fixer)	Sodium thiosulphate	$Na_2S_2O_3 \cdot 5H_2O$
Milk of magnesia	Magnesium hydroxide	$Mg(OH)_2$
Epsom salt	Magnesium sulphate heptahydrate	$MgSO_4 \cdot 7H_2O$
Glauber's salt	Sodium sulphate decahydrate	$Na_2SO_4 \cdot 10H_2O$
Borax	Sodium tetraborate decahydrate	$Na_2B_4O_7 \cdot 10H_2O$
Alum	Potassium aluminium sulphate	$KAl(SO_4)_2 \cdot 12H_2O$
Mohr's salt	Ferrous ammonium sulphate	$(NH_4)_2Fe(SO_4)_2 \cdot 6H_2O$
Blue vitriol	Copper sulphate pentahydrate	$CuSO_4 \cdot 5H_2O$
Green vitriol	Ferrous sulphate heptahydrate	$FeSO_4 \cdot 7H_2O$
White vitriol	Zinc sulphate heptahydrate	$ZnSO_4 \cdot 7H_2O$
Red lead	Lead tetroxide	Pb_3O_4
Litharge	Lead monoxide	PbO

Everyday Name	Chemical Name	Formula
Dry ice	Solid carbon dioxide	$CO_2(s)$
Heavy water	Deuterium oxide	D_2O
Vinegar	Dilute acetic acid	CH_3COOH (5–8%)
Spirit / Rectified spirit	Ethanol	C_2H_5OH
Wood spirit (poison)	Methanol	CH_3OH
Marsh gas	Methane	CH_4
Laughing gas	Nitrous oxide	N_2O
Tear gas	Chloroacetophenone / CS gas	Various
Mustard gas	Dichlorodiethyl sulphide	$(ClCH_2CH_2)_2S$
Aqua regia	3 HCl + 1 HNO_3	— (dissolves Au, Pt)
Aqua fortis	Concentrated nitric acid	HNO_3
Oil of vitriol	Concentrated sulphuric acid	H_2SO_4
Spirit of salt	Hydrochloric acid	HCl
Sugar of lead	Lead acetate	$Pb(CH_3COO)_2$
Vermillion / Sindoor	Mercuric sulphide	HgS
Galena	Lead sulphide	PbS
Cinnabar	Mercuric sulphide	HgS
Copperas	Ferrous sulphate	$FeSO_4 \cdot 7H_2O$
Producer gas	Carbon monoxide + Nitrogen	CO + N_2
Water gas	Carbon monoxide + Hydrogen	CO + H_2
Phosgene	Carbonyl chloride	$COCl_2$

× **COMMON TRAP**

Plaster of Paris vs Gypsum — a classic trap: Gypsum is $CaSO_4 \cdot 2H_2O$ (two water molecules). POP is $CaSO_4 \cdot \frac{1}{2}H_2O$ (half water). POP is made from gypsum by heating to 120–130 °C (loses 1.5 of 2 water molecules). When POP is mixed with water, it absorbs water and sets hard — converting back to gypsum. This is used in surgical casts, dental moulds, and sculptures.

PART V — ALLOYS AND METALLURGY

The Alloys Table — Priority Memorisation

Alloy composition questions appear in every SSC and UPSC paper. Learn the top 15 cold.

Alloy	Composition	Primary Use
Brass	Copper (60–80%) + Zinc	Utensils, decorative items, musical instruments
Bronze	Copper (~90%) + Tin (~10%)	Statues, coins, bells, ancient tools
Bell metal	Copper (78%) + Tin (22%)	Temple bells
Gun metal	Cu (88%) + Sn (10%) + Zn (2%)	Gun barrels, marine fittings
German silver	Cu (60%) + Zn (20%) + Ni (20%)	Tableware, resistance wires
Stainless steel	Fe (74%) + Cr (18%) + Ni (8%)	Kitchen utensils, surgical instruments
Mild steel	Fe + C (0.1–0.3%)	Construction, bridges
High-carbon steel	Fe + C (0.5–1.5%)	Cutting tools, blades
Cast iron	Fe + C (~3%)	Engine blocks, pipes
Wrought iron	Fe + C (<0.1%)	Ornamental gates, chains
Solder	Pb (50%) + Sn (50%)	Joining electrical wires
Duralumin	Al (95%) + Cu (4%) + Mg + Mn	Aircraft frames, lightweight structures
Magnalium	Al (95%) + Mg (5%)	Balance beams, optical instruments
Nichrome	Ni + Cr + Fe	Heating coil (high electrical resistance)
Alnico	Al + Ni + Co + Fe	Strong permanent magnets
Amalgam	Mercury + another metal	Dental fillings (Hg + Ag + Sn)
Invar	Fe + Ni (36%)	Precision measuring instruments (low thermal expansion)
22-carat gold	Gold 91.6% + Cu/Ag	Jewellery
18-carat gold	Gold 75% + Cu/Ag	Jewellery

Alloy	Composition	Primary Use
Sterling silver	Silver 92.5% + Cu 7.5%	Jewellery, coins

✔ **KEY FACT**

Brass = copper + zinc. Bronze = copper + tin. These two are confused in nearly every exam. Remember: **Brass = Bronze minus Tin plus Zinc**, or just: **Brass = BZ (Bronze with Zinc substituted)**. Bronze has tin; brass has zinc.

PART Y — FERTILISERS

Chapter Y1 — The NPK Macronutrients

Nutrient	Symbol	Role in Plant	Deficiency
Nitrogen	N	Leaf growth, chlorophyll production	Yellowing of older leaves (chlorosis)
Phosphorus	P	Root growth, flower and fruit development	Purple/reddish leaves, poor root system
Potassium	K	Water regulation, disease resistance, stem strength	Leaf-edge scorching (marginal necrosis)

Chapter Y2 — Common Nitrogenous Fertilisers

Fertiliser	Formula	% Nitrogen	Key fact
Urea	$CO(NH_2)_2$	46%	Highest nitrogen content of all solid fertilisers
Ammonium sulphate	$(NH_4)_2SO_4$	21%	Also provides sulphur
Ammonium nitrate	NH_4NO_3	33–34%	Also explosive
Calcium ammonium nitrate (CAN)	NH_4NO_3 + $CaCO_3$	25–26%	Safer form
Sodium nitrate (Chile saltpetre)	$NaNO_3$	16%	Natural source

Chapter Y3 — Biofertilisers

Organism	Type	Action
Rhizobium	Symbiotic (in legume root nodules)	Fixes atmospheric N_2 to NH_3
Azotobacter	Free-living in soil	Fixes atmospheric nitrogen
Azospirillum	Free-living	Nitrogen fixation; plant growth hormones
Anabaena, Nostoc (cyanobacteria)	Free-living	Nitrogen fixation (important in paddy fields)

Organism	Type	Action
Mycorrhiza (VAM fungi)	Symbiotic	Extends root uptake area; phosphate mobilisation
Pseudomonas, Bacillus	Free-living	Solubilise insoluble phosphate in soil

⚠ EXAM ALERT

Rhizobium bacteria live in root nodules of legumes (pulses — peas, beans, lentils, groundnuts). They convert atmospheric N_2 into ammonia (NH_3) through biological nitrogen fixation. This is why farmers rotate crops — growing legumes restores nitrogen to the soil without chemical fertilisers.

PART AA — INDUSTRIAL PROCESSES

The Process Table — Every Exam Uses This

Process	Product	Key conditions / catalyst
Haber process	Ammonia (NH_3)	$N_2 + 3H_2 \rightarrow 2NH_3$; Fe catalyst; 200 atm pressure; 450 °C
Ostwald process	Nitric acid (HNO_3)	Ammonia + air → NO → NO_2 → HNO_3; Pt-Rh catalyst
Contact process	Sulphuric acid (H_2SO_4)	$SO_2 + O_2 \rightarrow SO_3 \rightarrow H_2SO_4$; V_2O_5 catalyst; 450–550 °C
Solvay process	Washing soda (Na_2CO_3)	$NaCl + NH_3 + CO_2 + H_2O$
Castner-Kellner	Caustic soda (NaOH)	Electrolysis of NaCl brine
Hall-Héroult	Aluminium (Al)	Electrolysis of Al_2O_3 in molten cryolite; 950 °C
Bayer process	Alumina (Al_2O_3)	Bauxite + NaOH (leaching)
Bessemer process	Steel from pig iron	Oxygen blast through molten iron
Frasch process	Elemental sulphur	Superheated water + compressed air
Down's process	Sodium	Electrolysis of molten NaCl
Mond process	Pure nickel	Ni + CO → $Ni(CO)_4$ → heated → pure Ni
Zone refining	Ultra-pure semiconductors	Si, Ge, Ga — for electronics
Cracking	Petrol from heavier fractions	Thermal or catalytic; breaks long hydrocarbon chains
Reforming	High-octane petrol	Rearranges hydrocarbon molecules; Pt catalyst

◎ **MNEMONIC**

For the three big acid/base industrial processes: **"Haber makes ammonia (H for Hydrogen), Ostwald makes nitric (Ozone-killing NOx), Contact makes sulphuric (C for Caustic)"**. Catalyst clue: Haber = **Fe**; Contact = **V_2O_5** (Vanadium); Ostwald = **Pt** (Platinum).

PART AB — ORGANIC CHEMISTRY DEPTH

Chapter AB1 — IUPAC Naming Rules

1. Find the longest carbon chain containing the principal functional group → root name.
2. Number the chain from the end closer to the principal group.
3. Name substituents as prefixes (alphabetical order).
4. Principal group suffix (highest priority first in table below).

Functional Group	Suffix	Example
Carboxylic acid (−COOH)	-oic acid	Ethanoic acid (acetic acid)
Ester (−COO−)	-oate	Ethyl ethanoate
Aldehyde (−CHO)	-al	Ethanal (acetaldehyde)
Ketone (>C=O)	-one	Propan-2-one (acetone)
Alcohol (−OH)	-ol	Ethanol, methanol
Amine ($-NH_2$)	-amine	Methylamine
Alkene (C=C)	-ene	Ethene
Alkyne (C≡C)	-yne	Ethyne (acetylene)
Alkane (no double bond)	-ane	Methane, ethane

Chapter AB2 — Named Reactions (High-Frequency)

Reaction	What happens	Use
Haber	$N_2 + H_2 \rightarrow NH_3$	Ammonia for fertilisers
Saponification	Fat/oil + NaOH → soap + glycerol	Soap making
Esterification	Alcohol + acid → ester + water	Perfumes, solvents
Hydrogenation	Alkene + H_2 → alkane (Ni cat)	Vegetable oil → Vanaspati ghee
Fermentation	Sugar → ethanol + CO_2 (yeast)	Alcoholic beverages, bread
Markovnikov rule	H of HX adds to C with more H atoms	Alkene addition reactions
Wurtz reaction	2RX + 2Na → R-R + 2NaX	Synthesise higher alkanes
Aldol condensation	2 aldehydes (with α-H) → β-hydroxy aldehyde	C-C bond formation
Friedel-Crafts	Benzene + R-Cl ($AlCl_3$) → alkylbenzene	Aromatic alkylation

Reaction	What happens	Use
Sandmeyer	Aryl-N_2^+ + CuX → aryl-X	Aryl halide synthesis

Chapter AB3 — Tests for Functional Groups

Test	What it detects	What you observe
Tollens' (silver mirror)	Aldehyde	Silver mirror forms on test tube wall
Fehling's	Aldehyde / reducing sugar	Brick-red Cu_2O precipitate
Benedict's	Reducing sugar	Red/orange Cu_2O precipitate
Iodoform test	Methyl ketones + ethanol	Yellow CHI_3 (iodoform) precipitate
Lucas test	1°/2°/3° alcohols	Turbidity time: 3° immediate, 2° slow, 1° no reaction
Bromine water	Alkene/alkyne (unsaturation)	Brown colour disappears
Baeyer's ($KMnO_4$)	Alkene	Purple $KMnO_4$ → colourless
Sodium bicarbonate	Carboxylic acid	CO_2 effervescence (fizzing)
$FeCl_3$ (violet)	Phenol	Violet/purple colour
Carbylamine test	Primary amine	Foul-smelling isocyanide

PART AC — BIOMOLECULES

Carbohydrates

- **Monosaccharides (simple sugars):** Glucose ($C_6H_{12}O_6$), fructose, galactose. Cannot be hydrolysed further.
- **Disaccharides:** Two monosaccharides joined. Sucrose = glucose + fructose; Maltose = glucose + glucose; Lactose (milk sugar) = glucose + galactose.
- **Polysaccharides:** Long chains. Starch (α-glucose, plant energy storage), Glycogen (α-glucose, highly branched, animal energy storage in liver/muscles), Cellulose (β-glucose, plant cell walls — humans cannot digest).

⚠ EXAM ALERT

Sucrose is NOT a reducing sugar (the functional aldehyde/ketone groups are locked in the glycosidic bond). Maltose, lactose, and glucose ARE reducing sugars. This distinction tests Benedict's/Fehling's test results.

Proteins

- Built from **amino acids** linked by **peptide bonds**.
- 20 amino acids in proteins; 9 are essential (body cannot make them, must come from diet): Histidine, Isoleucine, Leucine, Lysine, Methionine, Phenylalanine, Threonine, Tryptophan, Valine.
- Structure levels: Primary (amino acid sequence) → Secondary (α-helix / β-sheet) → Tertiary (3D folded shape) → Quaternary (multiple subunits, e.g., haemoglobin).

Nucleic Acids

- **DNA** — double helix; deoxyribose sugar; base pairs A-T (2 H-bonds), G-C (3 H-bonds). Carries genetic information.
- **RNA** — single-stranded; ribose sugar; base pairs A-U (uracil replaces thymine), G-C.
- DNA replication is **semi-conservative** — each strand of the double helix is used as a template. Proved by Meselson and Stahl (1958).

Vitamins — The Full Table

Vitamin	Chemical Name	Solubility	Deficiency Disease
A	Retinol	Fat	Night blindness, xerophthalmia
B_1	Thiamine	Water	Beri-beri

Vitamin	Chemical Name	Solubility	Deficiency Disease
B_2	Riboflavin	Water	Cheilosis, angular stomatitis
B_3	Niacin (Nicotinic acid)	Water	Pellagra
B_5	Pantothenic acid	Water	Burning feet syndrome
B_6	Pyridoxine	Water	Anaemia, neuritis
B_7 / H	Biotin	Water	Hair loss, dermatitis
B_9	Folic acid	Water	Megaloblastic anaemia, neural tube defects
B_{12}	Cyanocobalamin	Water	Pernicious anaemia
C	Ascorbic acid	Water	Scurvy
D	Calciferol	Fat	Rickets (children), osteomalacia (adults)
E	Tocopherol	Fat	Sterility, RBC fragility
K	Phylloquinone	Fat	Excessive bleeding, poor clotting

◎ MNEMONIC

Fat-soluble vitamins: **A, D, E, K** — "Anita Drinks Every Kombucha." All other vitamins (the B family and C) are water-soluble. Only B_{12} comes exclusively from animal sources — strict vegans need supplements.

APPENDIX A — Master Mnemonics

Activity series (most → least reactive): K, Na, Ca, Mg, Al, Zn, Fe, Sn, Pb, [H], Cu, Hg, Ag, Au, Pt

"**P**lease **S**end **C**harlie's **M**oney **A**t **Z**ara **F**or **S**ome **P**ost — **H**e **C**an't **H**elp **S**pending **A**ll **P**ounds" (K=Potassium, Na=Sodium, Ca=Calcium, Mg=Magnesium, Al=Aluminium, Zn=Zinc, Fe=Iron/Ferrous, Sn=Tin, Pb=Lead, H=Hydrogen, Cu=Copper, Hg=Mercury/Hydrargyrum, Ag=Silver, Au=Gold/Aurum, Pt=Platinum)

Electronegativity order: F > O > N > Cl > Br > I "**F**rench **O**cean **N**avy **Cl**eans **Br**itish **I**slands"

Fat-soluble vitamins: A, D, E, K — "Anita Drinks Every Kombucha"

Alkane prefix order (1-10): Meth-, Eth-, Prop-, But-, Pent-, Hex-, Hept-, Oct-, Non-, Dec- "**M**y **E**lder **P**arrot **B**ites **P**eople; **H**e's **O**ld **N**ow, **D**oc"

Periodic table blocks: s (Groups 1–2), p (Groups 13–18), d (Groups 3–12), f (lanthanides + actinides) "**s**mart **p**eople **d**o **f**antastic work" — s, p, d, f

Noble gases: He, Ne, Ar, Kr, Xe, Rn — "**H**e **N**ever **A**rgued; **K**ept **Xe**rocopy **R**ight"

APPENDIX B — 25-Question Mini-Mock

Attempt without looking at the answers. If you score below 18, revisit the relevant Part.

1. What is the chemical name of bleaching powder?
2. Aqua regia is a mixture of which two acids and in what ratio?
3. Which metal is the best conductor of electricity?
4. What is the chemical formula of washing soda?
5. Name the process by which aluminium is extracted from bauxite.
6. What is the monomer of natural rubber?
7. Name the scientist who discovered penicillin and in what year.
8. Which vitamin deficiency causes scurvy?
9. What is the main component of biogas?
10. In the activity series, which metal is the most reactive?
11. Brass is an alloy of which two metals?
12. What is the pH range of human blood?
13. Which gas turns lime water milky?
14. What does BOD stand for and what does a high BOD indicate?
15. Which polymer is used in non-stick cookware?
16. What is the calorific value of hydrogen compared to petrol?
17. Name the industrial process that makes ammonia.
18. Vulcanisation of rubber was discovered by whom and in what year?
19. Which vitamin is the only one that comes exclusively from animal sources?
20. What type of bond is present in H_2O that explains its high boiling point?
21. Which element has the highest electronegativity?
22. Name the substance responsible for the "laughing gas" effect.
23. What is the main difference between soap and detergent in hard water?
24. Which disease is caused by mercury poisoning in water?
25. Name three substances that sublime.

Answers: 1. Calcium hypochlorite ($CaOCl_2$) 2. 3 parts HCl + 1 part HNO_3 3. Silver (Ag) — copper is used in wiring for economy 4. $Na_2CO_3 \cdot 10H_2O$ 5. Hall-Héroult process (electrolysis of Al_2O_3 in molten cryolite) 6. Isoprene (2-methylbuta-1,3-diene) 7. Alexander Fleming, 1928 8. Vitamin C (Ascorbic acid) 9. Methane (CH_4, ~60%) 10. Potassium (K) / Caesium (Cs) 11. Copper + Zinc 12. 7.35 to 7.45 13. Carbon dioxide (CO_2) 14. Biological Oxygen Demand; high BOD = more organic pollution 15. Teflon (PTFE) 16. Hydrogen: ~150,000 kJ/kg vs petrol: ~45,000 kJ/kg (about 3× higher) 17. Haber process 18. Charles Goodyear, 1839 19. Vitamin B_{12} (Cyanocobalamin) 20. Hydrogen bonding 21. Fluorine (F) 22. Nitrous oxide (N_2O) 23. Soap forms insoluble scum with Ca^{2+}/Mg^{2+} ions; detergent works normally in hard water 24. Minamata disease 25. Dry ice (CO_2), Iodine, Naphthalene (also camphor, ammonium chloride)

APPENDIX C — Trap Recognition Card

Keep these traps in mind during any chemistry question:

Trap	The wrong answer (popular mistake)	The right answer
Best conductor	Copper	Silver (Ag)
Liquid metal at room temp	Only mercury	Mercury; also gallium (Ga, melts at 29.8 °C)
Lightest metal	Aluminium	Lithium (Li)
First plastic	PVC or nylon	Bakelite (1907)
First synthetic fibre	Polyester	Nylon (1935)
Brass vs Bronze	Both are "copper + something"	Brass = Cu + Zinc; Bronze = Cu + Tin
POP vs Gypsum	Both are calcium sulphate	Gypsum = $CaSO_4{\cdot}2H_2O$; POP = $CaSO_4{\cdot}\frac{1}{2}H_2O$
GWP strongest gas	CO_2	SF_6 (23,500× CO_2)
Main greenhouse gas	Water vapour	For exams, CO_2 is the expected answer
Ozone protocol	Kyoto	Montreal (1987) targets CFCs
Laughing gas	CO or N_2	N_2O (nitrous oxide)
Mustard gas	Actual gas	Liquid at room temperature — an oily liquid, not a gas
Aqua regia ratio	1:3 or equal	3 HCl : 1 HNO_3 (by volume)
pH of blood	7.0 (neutral)	7.35–7.45 (slightly alkaline)
Sublimation examples	Only dry ice	Dry ice, iodine, naphthalene, camphor